



FINAL YEAR PROJECT · COMPUTER SCIENCE

VITAL GUARDIAN

AI-Powered Real-Time ICU Patient Monitoring System

The Problem

Global nursing shortages make continuous bedside observation unscalable — delayed responses to critical patient events cost lives.

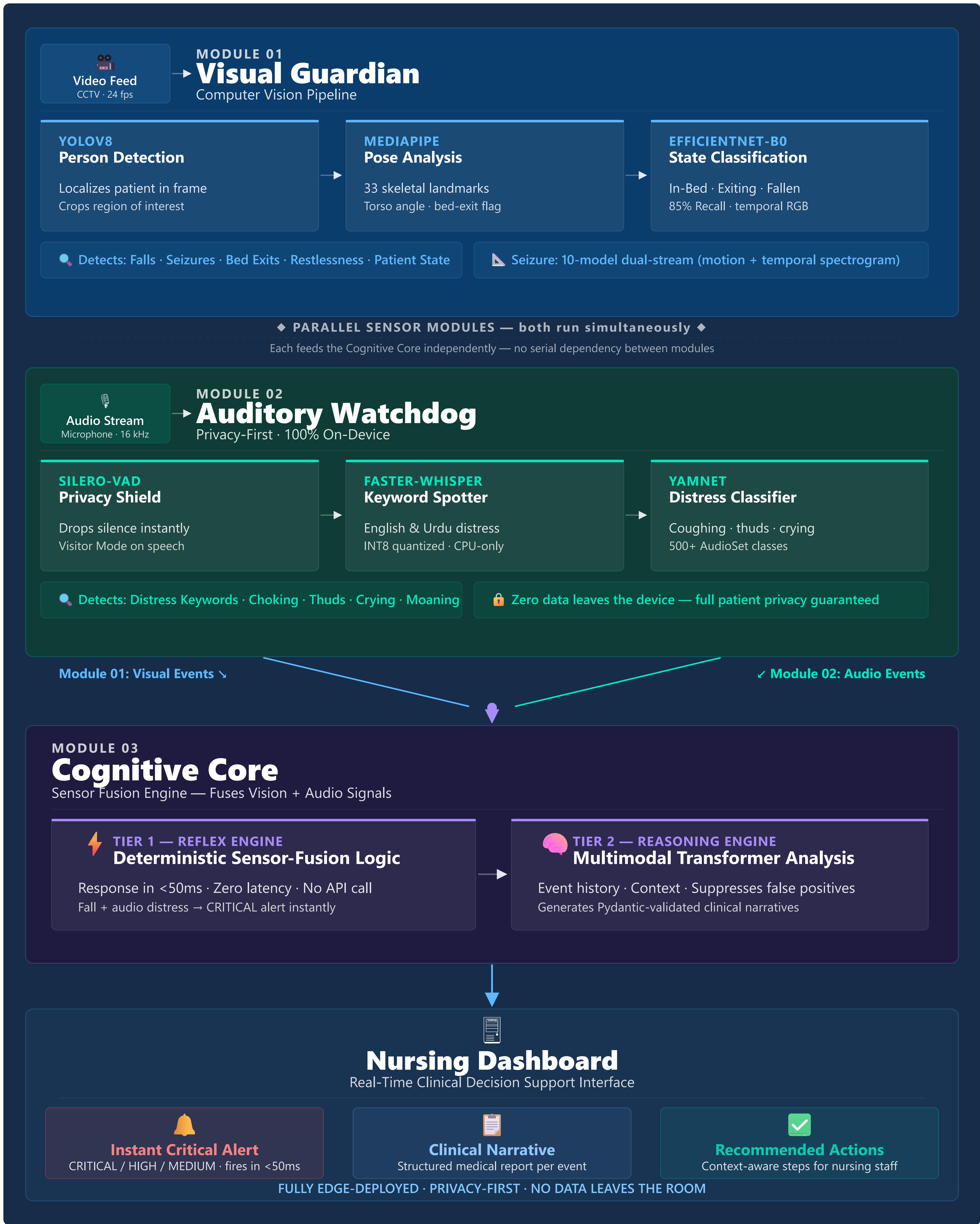
The Friction

Existing solutions rely on uncomfortable wearables or trigger excessive false alarms, causing dangerous alarm fatigue in clinical staff.

Our Objective

A continuous, non-invasive, multi-modal system with sub-50 ms critical event detection via AI sensor-fusion. No wearables. No raw data off-device.

SYSTEM ARCHITECTURE



CORE IMPACT

- Combating Alarm Fatigue**
Multi-modal corroboration eliminates false-positive single-sensor triggers.
- Staff Amplification**
A tireless digital co-attendant scaling oversight without staffing increases.
- Privacy by Design**
Fully edge-deployed. Zero raw audio or video ever leaves the room.
- Zero-Contact Monitoring**
No wearables. Complete patient comfort preserved at all times.

TECHNOLOGY STACK

